Mock B

110 min · 36 marks · 24 MCQ + 6 short answer · open book, no internet or AI. Circle one option per MCQ; tick **guessed** if unsure.

Section A: Multiple choice (1 mark each)

Question 1 guessed ☐ [1 mark]

Which one of the following would be the best use case for SaaS?

- **(a)** Applications with highly confidential and sensitive data
- **(b)** A mobile application
- **(c)** An application used by an organisation with people working around the world, where some have erratic or unreliable internet connections
- **(d)** Organisations looking to reduce on-premises IT infrastructure and capital expenditure
- **(e)** Applications that need a lot of user data

Question 2 guessed ☐ [1 mark]

Which one of the statements below about the number of workers in a MapReduce framework is correct?

- **(a)** Too many workers can reduce overall efficiency because it overloads the master node's task and increases data transfer overhead between workers
- **(b)** The number of workers has no effect on performance, because MapReduce automatically balances tasks perfectly
- **(c)** Using fewer workers than available CPUs is always better, because it reduces network usage and eliminates the risk of data skew
- **(d)** MapReduce performance depends only on the size of the input data, not on the number of workers assigned
- **(e)** Having too few workers is not efficient because total worker idle time increases

Question 3 guessed ☐ [1 mark]

Your users are reporting increased levels of latency. On checking you notice that CPU usage is actually low. What may be the primary cause then?

- **(a)** Servers are overprovisioned
- **(b)** Too many instances have been set up
- **(c)** Horizontal scaling
- **(d)** Network bottleneck
- **(e)** Usage of CDNs

Question 4 guessed ☐ [1 mark]

What is one key advantage of using virtual machines over containers?

- **(a)** Requires less memory space
- **(b)** More secure
- **(c)** Faster startup time
- **(d)** Easy to scale
- **(e)** More cost-effective

Question 5 guessed ☐ [1 mark]

A team clusters 1,000,000 customer records (each a vector of purchase attributes) into k = 5 groups using k-means written as a MapReduce job. During one round, what does the map step emit?

(a) One pair per customer record: (the ID of its nearest centroid, that customer's vector)

(b) One pair per cluster: (cluster ID, the new centroid and the number of points in it)

(c) One pair per customer record: (customer ID, its distance to each of the 5 centroids)

(d) One pair per map task: (cluster ID, the sum of the vectors that task has seen)

(e) Nothing: the map step only reads the data, and the reduce step assigns each point to a cluster

Question 6 guessed ☐ [1 mark]

You are running a cloud environment and realise you need a load balancer. What is its primary purpose?

(a) To ensure a high bandwidth between your application and the client

(b) To automatically adjust the necessary amount of computing resources that are used by your applications

(c) To distribute incoming service requests across multiple servers

(d) To ensure that data in the cloud is synchronised across multiple servers

(e) To manage the various computing resources within a datacenter

Question 7 guessed ☐ [1 mark]

A public-sector platform must keep both data and metadata within one country's borders and prevent foreign access under all circumstances. Which cloud deployment model is the best fit?

(a) Public Cloud

(b) Private Cloud

(c) Hybrid Cloud

(d) Sovereign Cloud

(e) Multi Cloud

Question 8 guessed ☐ [1 mark]

Which requirement is the clearest sign that a workload is a poor fit for FaaS?

(a) It is triggered by file uploads

(b) It requires long-lived stateful sessions and persistent in-process connections

(c) It performs short request-driven transformations

(d) It runs sporadically and scales by event volume

(e) It requires long-lived stateless sessions

Question 9 guessed ☐ [1 mark]

Which of the following scenarios is best suited for using a hybrid cloud model?

(a) A government agency hosting sensitive data

(b) A startup hosting a website

(c) A user hosting a blog

(d) A large enterprise with varying workloads

(e) A startup having a lot of data

Question 10 guessed ☐ [1 mark]

You realise that using containerised applications in a datacenter environment gives you certain advantages. Which one of the following could be one of them?

(a) Increases the physical size of the server infrastructure

(b) Eliminates the need for cloud-based applications

(c) Improves isolation between applications and operating system

(d) Reduces the number of physical servers needed

(e) Reduces the overall cost

Question 11 guessed ☐ [1 mark]

How does an Application Load Balancer enable reliability?

(a) By routing traffic to the least busy instances

(b) By routing traffic away from failed instances

(c) By caching frequently accessed content

(d) By replacing failed instances

(e) By routing traffic to new instances

Question 12 guessed ☐ [1 mark]

Who amongst the following are primary users of PaaS?

(a) Clients looking to use a software online

(b) Software developers

(c) System administrators

(d) Hardware engineers

(e) Network engineers

Question 13 guessed ☐ [1 mark]

A cloud provider runs the virtual machines of many customers on the same physical server. A program inside Customer X's VM is compromised and tries to read the memory that belongs to Customer Y's VM on that same server. Which component is mainly responsible for stopping this?

(a) The virtual private cloud's network access rules

(b) The hypervisor, which logically separates the VMs on the server

(c) The elastic load balancer in front of the servers

(d) The multitenant application's separation of each customer's data

(e) The cloud broker that sits between customers and the provider

Question 14 guessed ☐ [1 mark]

Which one of the following is an optimal strategy to employ when you have a very large number of reads and write operations to a scalable cloud database?

(a) Horizontal scaling

(b) Vertical scaling

(c) Increasing storage

(d) Cloud bursting

(e) Migrate to a faster database

Question 15 guessed ☐ [1 mark]

Out of the following five compute models, which one requires the most maintenance?

(a) PaaS and VMs

(b) FaaS and Containers

(c) PaaS and Containers

(d) Containers and VMs

(e) PaaS and FaaS

Question 16 guessed ☐ [1 mark]

You have an idea for an app that finds local festivity celebration parties. During the few festivities throughout the year (e.g. New Year, National Day, etc.) many users will use your app to find the most suitable party for them to go to based on users' criteria. Which AWS service is the most suitable to the usage profile and also cost effective?

(a) EC2: Virtual server to run your app

(b) S3: Store static app contents (party flyers, images) and serve directly to users

(c) Elastic Beanstalk: Managed EC2 with auto-scaling

(d) CloudFront: Cache contents and reduce latency for app users

(e) Lambda: App logic as a serverless function

Question 17 guessed ☐ [1 mark]

Which of the following is an example of Software as a Service (SaaS)?

(a) Microsoft Azure

(b) Amazon EC2

(c) Google Cloud Platform

(d) Google Photos

(e) Amazon EMR

Question 18 guessed ☐ [1 mark]

Which one of the following statements is correct regarding virtualization?

(a) Virtualization increases compatibility among hardware that use different standards

(b) Operating System virtualization is better suited for an application which requires direct communication with hardware resources

(c) In hardware-assisted virtualization, the guest operating system directly communicates with hardware resources

(d) Virtualization technology allows the isolation of applications running on the same server

(e) Virtualization allows multitenancy only when it is Type 2

Question 19 guessed ☐ [1 mark]

Assuming you are moving from an on-premise deployment to a cloud deployment, which is the most important area of concern?

(a) Scalability

(b) Security

(c) Load Balancing

(d) Storage

(e) Bandwidth

Question 20 guessed ☐ [1 mark]

A company has already made significant investments in on-premise infrastructure. Migrating to the cloud would require substantial additional costs for integration. Which of the following is the most appropriate conclusion based on this scenario?

(a) Cloud is always cheaper because it reduces upfront capital expenditure

(b) Reserved cloud pricing eliminates migration and integration costs

(c) On-premise solutions always have lower operational costs than cloud

(d) Past investments should not be considered when evaluating future decisions

(e) Cloud adoption may be less attractive due to high migration and integration costs

Question 21 guessed ☐ [1 mark]

A video-sharing site runs its upload web servers and its video-converting (encoding) software on the same auto-scaling group of servers. After one video goes viral, many long conversions start at once. Uploads begin to time out even though there is plenty of storage space. Which change most directly fixes this?

(a) Increase the size of the storage bucket that holds the videos

(b) Put a content delivery network in front of the upload servers

(c) Run the encoders in their own auto-scaling group; the upload servers save the raw video and leave a job message in a queue, and the encoders pick jobs from that queue

(d) Have each upload server call an encoder directly and wait for the conversion to finish before replying to the user

(e) Replace the servers with one larger server type for both jobs

Question 22 guessed ☐ [1 mark]

You wish to migrate to the cloud to run a simple server side computation which maintains no state about clients. Which of the following is best suited for this purpose?

(a) Amazon EC2

(b) REST

(c) SOAP

(d) AWS Lambda

(e) Amazon EMR

Question 23 guessed ☐ [1 mark]

Which of the following best describes a serverless application?

(a) A serverless application is usually developed in the context of PaaS

(b) The developer of a serverless application needs to pay a small fee for the idle periods

(c) A serverless application does not need a server/host to run on

(d) The developer of a serverless application does not need to provision or manage any servers

(e) The developer of a serverless application has lower costs

Question 24 guessed ☐ [1 mark]

According to the MapReduce execution flow described in the slides, which statement most accurately describes the data flow and the division of responsibility between the developer and the system?

- **(a)** The developer codes the Map function, which writes intermediate data to a shared storage server, and the Reduce function, which reads from that shared storage to aggregate results
- **(b)** The developer codes the Map and Reduce functions, but must also explicitly implement the "Shuffle and Sort" logic to ensure intermediate data is correctly grouped for the reducers
- **(c)** The Master instance partitions the input data, assigns segments to Map instances, and then directly sends the intermediate output from each Map instance to the correct Reduce instance
- **(d)** The developer's Map function writes intermediate results to the local disk of its worker node; the MapReduce system is then responsible for shuffling, merging, and sorting this data before feeding it to the developer's Reduce function
- **(e)** Map instances run in parallel, and Reduce instances run in parallel. To optimize performance, a Reduce worker is always placed on the same node as the Map worker that generated its data, eliminating network shuffling

Section B: Short answer (2 marks each)

Question 25 [2 marks]

Your company is using a public cloud environment but is facing some security concerns regarding its data. As the lead cloud engineer, you are asked to come up with the solution, while keeping costs down. How would you do it? Explain your answer.

Question 26 [2 marks]

AWS provides instances that are optimized for different workloads. Some instances are memory-optimized, others are I/O optimized. You are assigned a project to be run on AWS. The project requires 60 Peta FLOPs and 800,000 I/O operations to complete. Assume that compute and I/O operation are sequential. The details of each instance are provided below. Which instance should you choose to optimize your project cost? Please show your working.

Tera = 10^12. Peta = 10^15. FLOP = Floating Point Operations. ms = 10^-3 seconds.

Name	Computation speed (TeraFLOP/s)	I/O latency (ms/IOop)	Price ($/sec)
m7 (General purpose)	1	3	1
i8 (I/O optimized)	1	1	2
c5a (Compute optimized)	3	2	2
r7 (Memory optimized)	1	3	2
g4 (GPU)	30	10	5

Question 27 [2 marks]

A ticketing company sells concert tickets. Traffic is close to zero on most days, but when a popular concert goes on sale, demand jumps to about 200 times normal for roughly 30 minutes. The company's on-premise servers are sized for the peak, sit idle the rest of the time, and still crash on sale days. Propose two ways the company can cope with the spike without paying for idle capacity afterwards.

Question 28 [2 marks]

While setting up a new cloud deployment, you are tasked with ensuring that data availability must be high and that you must also comply with data protection guidelines which mandate that you must store Singapore customer data only within Singapore. How can you ensure compliance in such a scenario while keeping costs down? Briefly explain your answer.

Question 29 [2 marks]

A bank wants to move away from its legacy application and modernise it with strict compliance and some sensitive data. Which cloud service and cloud deployment model would you use and why?

Question 30 [2 marks]

You are responsible for the cloud infrastructure in company WeAreAwesome Inc. As awesome as you are, you have however noticed that your VMs that are running on the cloud are costing more than they should. Given your understanding of Cloud Computing, you wish to optimise the cost of running the VMs. Give two ways in which you can achieve this.